

Challenge (Habanero)

- Q1.** Two hikers are traveling on parallel trails. The slope of the first trail is 2. What is the slope of the second trail?
- (A) 2
(B) -2
(C) 1
(D) 0
(E) $\frac{1}{2}$
- Q2.** A park ranger is monitoring the population of deer in a nature reserve. The equation $y = 3x + 2$ represents the population of deer over time. If a parallel line represents the population of rabbits, which equation could represent it?
- (A) $y = 3x - 2$
(B) $y = 2x + 1$
(C) $y = -3x + 1$
(D) $y = 4x + 2$
(E) $y = x + 3$
- Q3.** A group of friends are planning a road trip. They want to travel on parallel roads to minimize traffic. If the slope of the first road is $-\frac{1}{2}$, what is the slope of the second road?
- (A) $-\frac{1}{2}$
(B) $\frac{1}{2}$
(C) 2
(D) -2
(E) 1
- Q4.** A wildlife biologist is studying the migration patterns of birds. The equation $y = -2x + 5$ represents the migration pattern of one species. If a parallel line represents the migration pattern of another species, which equation could represent it?
- (A) $y = -2x - 3$
(B) $y = 2x + 1$
(C) $y = -3x + 2$
(D) $y = x - 2$
(E) $y = -x + 1$
- Q5.** A hiker is traveling on a trail with a slope of $\frac{3}{4}$. If a parallel trail has a y -intercept of 2, which equation could represent it?
- (A) $y = \frac{3}{4}x + 2$
(B) $y = \frac{3}{4}x - 1$
(C) $y = -\frac{3}{4}x + 2$
(D) $y = \frac{1}{2}x + 1$
(E) $y = \frac{2}{3}x + 3$
- Q6.** A conservationist is monitoring the population of animals in a nature reserve. The equation $y = 2x - 1$ represents the population of one species. If a parallel line represents the population of another species, which equation could represent it?
- (A) $y = 2x + 2$
(B) $y = -2x + 1$
(C) $y = 2x - 3$
(D) $y = x + 1$
(E) $y = -x + 2$
- Q7.** A group of friends are planning a camping trip. They want to travel on parallel roads to minimize traffic. If the slope of the first road is 1, what is the slope of the second road?
- (A) 1
(B) -1
(C) 2
(D) -2
(E) 0
- Q8.** A park ranger is monitoring the fuel consumption of vehicles in a nature reserve. The equation $y = -\frac{1}{3}x + 4$ represents the fuel consumption of one vehicle. If a parallel line represents the fuel consumption of another vehicle, which equation could represent it?
- (A) $y = -\frac{1}{3}x + 2$
(B) $y = \frac{1}{3}x + 1$
(C) $y = -\frac{1}{3}x - 1$
(D) $y = \frac{2}{3}x + 2$
(E) $y = -\frac{2}{3}x + 1$

Open Ended Questions (FRQ)

- Q9.** A wildlife biologist is studying the migration patterns of birds. The equation $y = 2x + 1$ represents the migration pattern of one species. If a parallel line represents the migration pattern of another species, find the equation of the parallel line that passes through the point $(2, 5)$.
- Q10.** A group of friends are planning a road trip. They want to travel on parallel roads to minimize traffic. If the slope of the first road is -2 , find the equation of the parallel road that passes through the point $(-1, 3)$.

Chef's Tips

- 1
- Tip 1: Emphasize the importance of understanding the concept of parallel lines and their slopes.
 - Tip 2: Encourage students to use real-world examples to illustrate the concept of parallel lines.
 - Tip 3: Remind students to check their work carefully to ensure accuracy.